ChE 210 Final Exam Name: Nick Labree

Wednesday 6 May 2026

SPECIAL NOTE: To receive full credit, you must show all of your work and write your final answers in the spaces provided in this booklet.

Standards of Conduct- Academic Integrity

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all your academic endeavors.

Cheating on Examinations: Either intentionally using or attempting to use unauthorized materials, information, people, artificial intelligence, or study aids; providing to, or receiving from another person, any kind of unauthorized assistance on any examination.

Please note a new Chemical Engineering department policy: Cell phone, electronic eyeglasses or other type of electronic device usage are strictly prohibited during examinations.

The above behaviors will not be tolerated and will result at a minimum in a zero on the exam or assignment. Other possible actions include an "F" on the final grade, investigation, and disciplinary action by the Office of the Dean of Students and in egregious cases, expulsion from UIC.

1	15
2	38
T	53

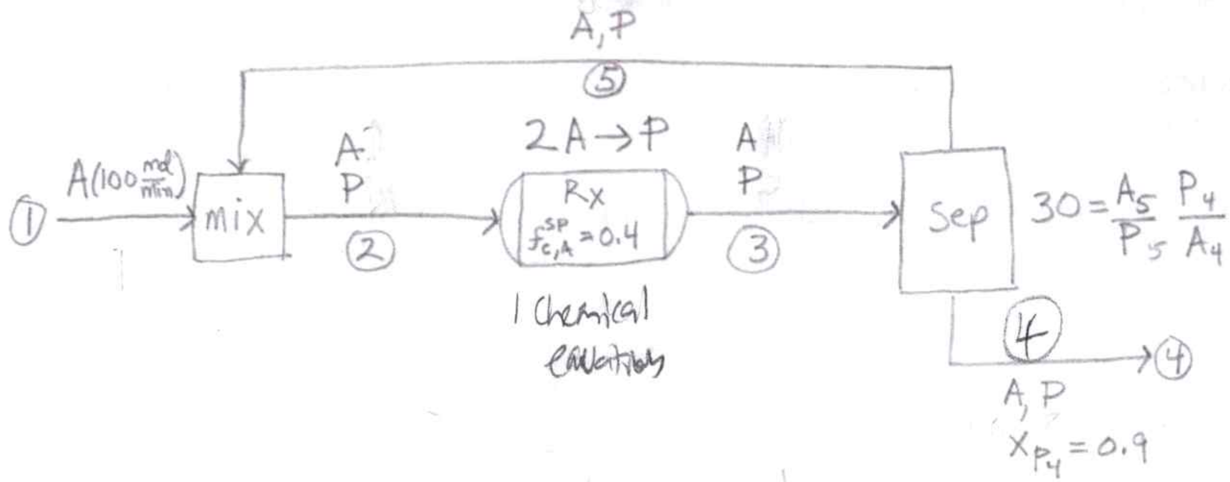
(1) (50 pts.) Consider the block flow diagram below. Notice that the streams are labeled 1 through 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains only species A. This stream is mixed with a recycle stream containing A and P where P is the reaction product. The mixture enters a reactor in which one reaction takes place:

$$2A \rightarrow P$$

The reactor converts 40% of A entering the reactor. The outlet stream from the reactor goes to a separator. Both streams leaving the separator are in equilibrium containing some A and P such that:

$$\alpha_{AB} = \frac{A_5}{P_5}\frac{P_4}{A_4}$$

where the separation factor α_{AB}= 30. Stream 5 is the stream enriched in A and is recycled back to the reactor. Stream 4 is enriched in the product P such that it is 90 mol% P. The fresh feed has species flow rates of A_1 = 100 mol A/min.



(a) Use a degree of freedom analysis to determine whether the problem as stated is well posed. (5 pts.)

4a. 9 components total   A (1)   AP (2)   AP (3)   AP (4)   AP (5)

2

4b. 1

4: 10

5a, 1 flow ($100 \frac{mol}{min}$)   1 2 3 4 5

5b. ~~4~~   0 1 1 1 1

5c, ~~3~~. $f_{c,A}^{sp} = 0.4$, $\alpha_{AB} = \frac{A_5, P_4}{P_5 A_4}$, $\alpha_{AB} = 30$

5d. ~~2~~   P & A

DoF = 0   5: 10   10 − 10 = 0

(Problem 1 Continued)

(b) Find all of the species molar flow rates not given in the problem statement. Fill in the table below with the appropriate species flow rates in mole/min to receive full credit. (40 pts.)

$A_3 = A_2 \cdot .6$?

$P_3 = \frac{1}{2}(A_2 \cdot .4) + P_2$?

$P_2 = P_5$

$A_2 = A_1 + A_5$

(100 mol/min)

$30 = \frac{A_5}{P_5} \cdot \frac{P_4}{A_4}$

$A_5 = A_3 - A_4$

$\frac{A_5 \cdot P_4}{P_5 \cdot (P_4 \cdot .1)}$

$P_5 = P_3 - P_4$ $\quad A_4 = P_4 \cdot .1$

$P_4 = P_3 - P_5$ $\quad A_4 = A_3 - A_5$

$30 =$

$A_2 = 100 + A_5$ $\quad A_2 - A_5 = 100$

$A_5 = A_2 - 100 \frac{mol}{min}$ $\quad -A_5 = 100 - A_2$

$\frac{30(P_5 \cdot A_4)}{P_4} = A_2 - 100$ $\quad A_5 = A_2 - 100$

$\frac{30(P_5 \cdot A_4)}{A_4 + (A_4 \cdot .10)}$

ratio between P4 & A4

$\frac{P_4}{.1P_4} = \frac{P_4}{A_4} = 10$

$\left(\frac{A_5}{P_5} \cdot \frac{P_4}{A_4}\right)$

$\frac{.6A_2}{(.2 A_2 + P_2)}$ $\quad \frac{A_2 \cdot .6}{\frac{1}{2}(A_2 \cdot .4) + P_2}$

$\frac{A_5}{P_5} \cdot 10 = 30$

$\frac{A_5}{P_5} = 3$

$A_5 = 3P_5$

$\frac{3A_2}{P_2} = 30$

$A_2 = 10P_2$

$A_2 = 100 + A_5$

$\frac{300P_5}{P_5}$ $\quad 3\frac{P_5}{P_5} \cdot \frac{P_4}{A_4} = 30$

$\frac{P_4}{A_4} = 10$ $\quad P_4 = 1$

$A_4 = .1$

if $A_4 = P_4 \cdot .1$ then

(Problem 1 Continued)

$A_3 = A_2 \cdot .6$

$P_3 = \frac{1}{2}(A_2 \cdot .4) + P_2$

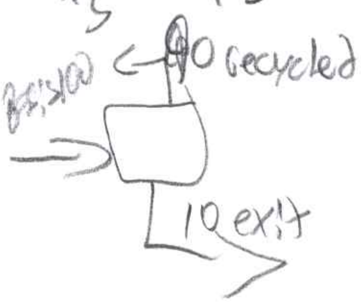
$P_2 = P_5$

$A_2 = A_1 + A_5$

$P_5 = P_3 - 1$

$.1 = A_3 - A_5$

$A_5 = 3P_5$



12

$A_5 + P_5 = 90$

$4P_5 = 90$

$P_5 = 22.5$

$= 67.5$

$100 + 67.5 = A_2$

$A_3 = 167.5 \cdot .6$

$P_3 = \frac{1}{2}(167.5 \cdot .4) + 22.5$

$= 56$

Species↓/Steam→	1	2	3	4	5
A	100	167.5 mol/s	100.5 mol/s	.1 mol/s	67.5 mol/s
P	----	22.5 mol/s	56 mol/s	1 mol/s	22.5 mol/s

(Problem 1 Continued)

(c) Calculate the overall fractional conversion based on species A (5 pts.)

$$\frac{\text{moles of reactant consumed}}{\text{moles fed}} = \cancel{0.4}$$

~~given?~~

$f_{c,A}^{ov} =$ ~~0.40298~~

$$\frac{67.5}{167.5} = 0.4$$

1

$2(P_3) + P_2$ = moles reactant consumed

15

(2) (50 pts.) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol (C_2H_5OH).

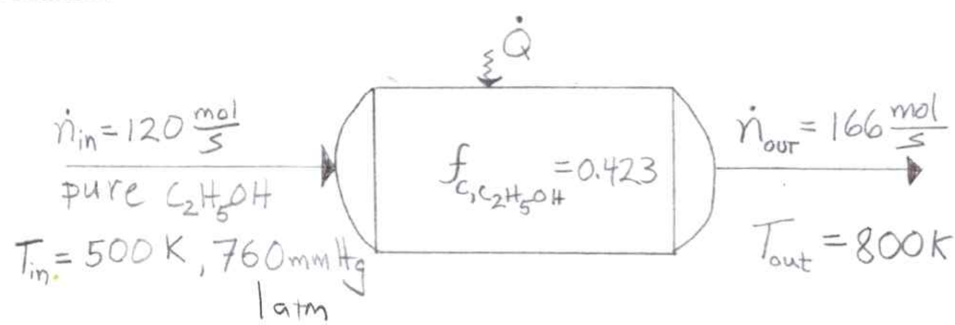
$$C_2H_5OH \rightarrow CH_3CHO + H_2 \qquad E \rightarrow A + H \qquad (R1)$$

An undesired side reaction produces ethyl acetate ($CH_3COOC_2H_5$)

$$2\ C_2H_5OH \rightarrow CH_3COOC_2H_5 + 2\ H_2 \qquad 2E \rightarrow EA + 2H \qquad (R2)$$

No accumulation or consumption

This process is carried out in a single reactor operating at steady state. The reactor has a feed rate of 120 mol/s of pure ethanol at 500K and 760 mm Hg. The reactor total product flow rate is 166 mol/s at 800K. The reactor converts 42.3% of the ethanol.



Data: The heat capacity in J/mol°C is given by $C_P = A + Bt$ where $t = T(K)/1000$. The table below gives the constants A and B.

Species	A	B
C_2H_5OH ✓	32.3	120
CH_3CHO	30.7	88.2
$CH_3COOC_2H_5$	60.9	191
H_2	28.4	1.54

(a) Calculate the all the species molar flow rates for the product stream in mol/s and write them into the Input/Output Table (see next page). Also, calculate the two extents of reaction, $\dot{\xi}_1$ and $\dot{\xi}_2$. (20 pts.)

120 mol/s · .423 = 50,76 product from reaction 1 & 2

C_2H_5OH

C_2H_5OH @ 500K
$32.3 + 0.5 \cdot 120 = 92,3$

$n_{out\,C_2H_5OH} = n_{in\,C_2H_5OH} - \dot{\xi}_1 - \dot{\xi}_2 = 6$

$\frac{1}{3}$ & $\frac{2}{3}$

$n_{out\,CH_3CHO} = \dot{\xi}_1 \cdot .5 = \cancel{8.46}\ \cancel{25,38}$ 8.46

$\dot{\xi}_1$ 16.92 $\dot{\xi}_2$ 33.84

$n_{out\,H_2} = (\dot{\xi}_1 \cdot .5) + (\dot{\xi}_2 \cdot \frac{2}{3}) = 31,02$

8

$\left[32.3x + \frac{120x^2}{2}\right]_{.298}^{.5} = 16.196$

$n_{out\,CH_3COOC_2H_5} = \dot{\xi}_2 \cdot \frac{1}{3} = 11.28$

$\left[32.3x + \frac{120x^2}{2}\right]_{0.298}^{0.8} = 49,286$

$\dot{\xi}_1 =$ ~~16.92~~ mol/s $\dot{\xi}_2 =$ ~~33.84~~ mol/s

(Problem 2 Continued)

(b) Finish filling enthalpy values into the Input/Output Table below. Use a reference state of $T_{ref} = 25°C$. (10 pts.)

	A	B	T(k)/1000
CH_3CHO	30.7	88.2	800k
$CH_3COOC_2H_5$	60.9	191	
H_2	28.4	1.54	

$$\int_{298k}^{800k} A+Bt\, dt = At + \frac{Bt^2}{2} \text{ for each of these}$$

$$\left[30.7x + \frac{88.2x^2}{2}\right]_{0.298}^{0.8}$$

$$(24.56 + 28.224) - (1.1026 + 3.916) = 47.76$$

$$\left[60.9x + \frac{191}{2}x^2\right]_{0.298}^{0.8}$$

$$(48.72 + 61.12) - (18.14 + 8.48) = 83.21$$

$$\left[28.4x + \frac{1.54x^2}{2}\right]_{0.298}^{.80}$$

$$(22.72 + 0.4928) - (8.4632 + 0.0683) = 14.68$$

$(166-120)+(120-\dot{\xi}_1-\dot{\xi}_2)$

10

Species	$\dot{n}_{in}$ (mol/s)	$\hat{H}_{in}$ (kJ/mol)	$\dot{n}_{out}$ (mol/s)	$\hat{H}_{out}$ (kJ/mol)
C_2H_5OH	120	16.196 kJ/mol	115.24	49.28636
CH_3CHO	----	----	8.46	47.75
$CH_3COOC_2H_5$	----	----	11.28	83.219
H_2	----	----	31.02	14.68

(Problem 2 Continued)

(c) Calculate the heat rate in kJ/s for the reactor. Indicate clearly whether it is heat in or out of the reactor. (20 pts.)

reactant $\hat{H}$ is heat coming in

product $\hat{H}_{out}$ is heat leaving

$$\sum \dot{n}_i \hat{H}_{in} - \sum \dot{n}_e \hat{H}_{out}$$

$$[(16.196 \cdot 120)] - [(115.24 \cdot 49.286) + (8.46 \cdot 47.72) + (11.28 \cdot 83.219) + (31.02 \cdot 14.68)]$$

$= -5537.47$ kJ/s

or

5537.47 kJ/s heat flowing out of the reactor

20

$\dot{Q}$ = 5537.47 kJ/s

38